Activity-2
CSA-1614-DWDM

Rellu Rohith Manikanta
192425326

1. Create a Sales Data Warehouse

G8

	A	B	C	D	E
1	Product	City	Month	Quantity	Sales Amount
2	Laptop	Chennai	January	5	₹2,50,000
3	Mobile Phone	Bengaluru	January	12	₹1,80,000
4	Headphones	Hyderabad	February	20	₹40,000
5	Monitor	Chennai	February	8	₹96,000
6	Keyboard	Mumbai	March	15	₹22,500
7	Laptop	Hyderabad	March	3	₹1,50,000
8	Mobile Phone	Chennai	April	10	₹1,50,000
9	Monitor	Bengaluru	April	6	₹72,000
10	Headphones	Mumbai	May	25	₹50,000
11	Keyboard	Hyderabad	May	18	₹27,000
12					
13					

2.fact table and dimension tables

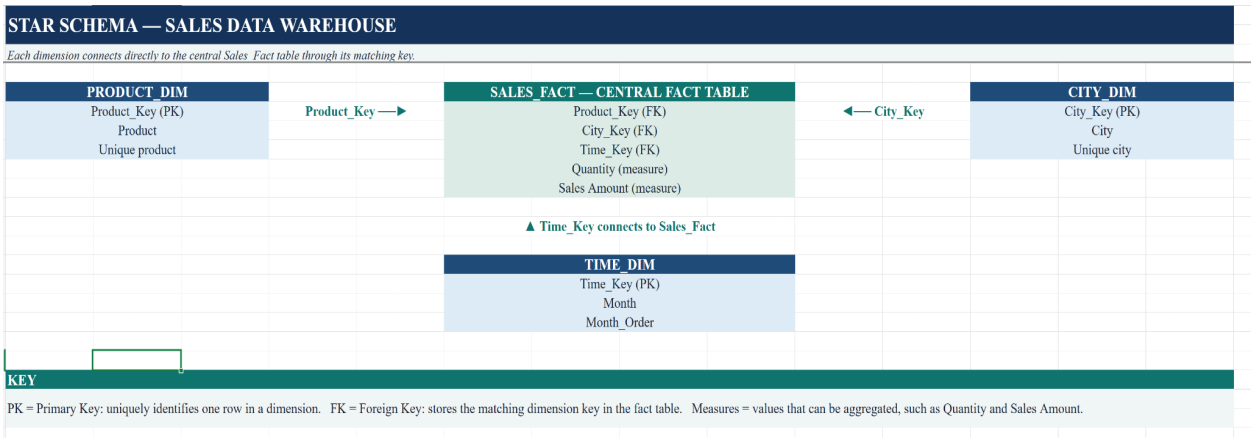
Fact table — Sales_Fact: stores the measurable business data, such as Quantity and Sales Amount.

Product dimension — Product_Dim: describes the products sold.

City dimension — City_Dim: describes the locations where sales occurred.

Time dimension — Time_Dim: describes the month of each sale.

3.star schema.



STAR SCHEMA — SALES DATA WAREHOUSE

Each dimension connects directly to the central Sales Fact table through its matching key.

PRODUCT_DIM		SALES_FACT — CENTRAL FACT TABLE		CITY_DIM
Product_Key (PK)	Product_Key ──▶	Product_Key (FK)	◀── City_Key	City_Key (PK)
Product		City_Key (FK)		City
Unique product		Time_Key (FK)		Unique city
		Quantity (measure)		
		Sales Amount (measure)		

▲ Time_Key connects to Sales_Fact

TIME_DIM
Time_Key (PK)
Month
Month_Order

KEY

PK = Primary Key: uniquely identifies one row in a dimension. FK = Foreign Key: stores the matching dimension key in the fact table. Measures = values that can be aggregated, such as Quantity and Sales Amount.

4.two questions the warehouse can answer

Which product generated the highest total sales amount?

Which city generated the highest sales amount in each month?

QUESTION 1 — TOTAL SALES BY PRODUCT

Product	Total Quantity	Total Sales Amount
Laptop	8	₹400,000
Mobile Phone	22	₹330,000
Headphones	45	₹90,000
Monitor	14	₹168,000
Keyboard	33	₹49,500

Highest sales product	Laptop

QUESTION 2 — WHICH CITY LEADS EACH MONTH?

Month	Chennai	Bengaluru	Hyderabad	Mumbai	Highest Sales City	Highest Sales Amount
January	₹250,000	₹180,000	₹0	₹0	Chennai	₹250,000
February	₹96,000	₹0	₹40,000	₹0	Chennai	₹96,000
March	₹0	₹0	₹150,000	₹22,500	Hyderabad	₹150,000
April	₹150,000	₹72,000	₹0	₹0	Chennai	₹150,000
May	₹0	₹0	₹27,000	₹50,000	Mumbai	₹50,000

2. Data Cleaning Challenge

1.Raw Data

	A	B	C	D	E	F
1	Customer_	Customer_Nai	Email	City	Age	
2	C001	Ananya Rao	ananya.rao@example.com	Chennai	24	
3	C002	Rahul Sharma	rahul.sharma@example.com	Bengaluru	31	
4	C003	Meera Iyer	meera.iyer@example.com	Hyderabad	28	
5	C004	Karthik Kumar	karthik.kumar@example.com	Chenai	35	
6	C005	Priya Nair		Chennai	26	
7	C006	Arjun Das	ARJUN.DAS@EXAMPLE.COM	Bangaluru	29	
8	C003	Meera Iyer	meera.iyer@example.com	Hyderabad	28	
9	C008	Divya Menon	divya.menon@example.com		32	
10	C009	Vikram Patel	vikram.patel@example.com	Mumbai	41	
11	C010	Sana Khan	sana.khan@example.com	Chennai	23	
12						
13						
14						

2. Cleaned data

	A	B	C	D	E	F
1	Customer_	Customer_Nai	Email	City	Age	
2	C001	Ananya Rao	ananya.rao@example.com	Chennai	24	
3	C002	Rahul Sharma	rahul.sharma@example.com	Bengaluru	31	
4	C003	Meera Iyer	meera.iyer@example.com	Hyderabad	28	
5	C004	Karthik Kumar	karthik.kumar@example.com	Chennai	35	
6	C005	Priya Nair	Not provided	Chennai	26	
7	C006	Arjun Das	arjun.das@example.com	Bengaluru	29	
8	C008	Divya Menon	divya.menon@example.com	Not provided	32	
9	C009	Vikram Patel	vikram.patel@example.com	Mumbai	41	
10	C010	Sana Khan	sana.khan@example.com	Chennai	23	
11						
12						
13						
14						

Processed

Step	Action	What was done
1	Identify data quality problems	Found 1 duplicate record, 2 missing values, spelling errors in city names, and inconsistent email spacing/capitalization.
2	Remove duplicates	Removed the repeated C003 record. Raw_Data has 10 records; Cleaned_Data has 9 unique records.
3	Handle missing values	Replaced the missing email for C005 and missing city for C008 with 'Not provided' so the records are retained.
4	Correct spelling errors	Changed 'Chenai' to 'Chennai' and 'Bangaluru' to 'Bengaluru'.
5	Standardize text	Trimmed spaces and converted the email for C006 to lowercase.
6	Verify corrected dataset	Checked that Cleaned_Data has no duplicate rows, no blank cells in the supplied fields, and standardized city names.